Solar RRL
www.solar-rrl.com

Modification of Hydrophobic Self-Assembled Monolayers with Nanoparticles for Improved Wettability and Enhanced Carrier Lifetimes Over Large Areas in Perovskite Solar Cells

*W. Hashini K. Perera, Mateus G. Masteghin, Hongjae Shim, Joshua D. Davies, Joshua L. Ryan, Steven J. Hinder, Jae S. Yun, Wei Zhang, K. D. G. Imalka Jayawardena, and S. Ravi P. Silva**

The development of perovskite solar cells (PSCs) with low recombination losses at low processing temperatures is an area of growing research interest as it enables compatibility with roll-to-roll processing on flexible substrates as well as with tandem solar cells. The inverted or p–i–n device architecture has emerged as the most promising PSC configuration due to the possibility of using low-temperature processable organic hole-transport layers and more recently, self-assembled monolayers such as [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz). However, devices incorporating these interlayers suffer from poor wettability of the precursor leading to pin hole formation and poor device yield. Herein, the use of alumina nanoparticles (Al_2O_3 nanoparticles (NPs)) for pinning the perovskite precursor on Me-4PACz is demonstrated, thereby improving the device yield. While similar wettability enhancements can also be achieved by using poly[(9,9-bis(3′-((*N*,*N*-dimethyl)-*N*-ethylammonium)-propyl)-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene)]dibromide (PFN-Br), a widely employed surface modifier, the incorporation of Al_2O_3 NPs results in significantly enhanced Shockley–Read–Hall recombination lifetimes exceeding 3 μs, which is higher than those on films coated directly on Me-4PACz and on PFN-Br-modified Me-4PACz. This translates to a champion power conversion efficiency of 19.9% for PSCs fabricated on Me-4PACz modified with Al_2O_3, which is a ≈20% improvement compared to the champion device fabricated on PFN-Br-modified Me-4PACz.

1. Introduction

Achieving net zero carbon emissions has been identified not only as the route toward negating the detrimental impact of climate change, but also as the route for realizing a more sustainable future for humanity.[1] Of the renewable energies expected to contribute toward achieving this target, photovoltaics (PVs) is anticipated to play a major role.[2] Among the existing PV technologies, perovskite solar cells (PSCs) have emerged as a promising candidate for future large-scale deployment owing to their high efficiencies and low fabrication cost.[3] Rapid developments in interlayers, passivation strategies, and bandgap optimization have allowed the realization of a record power conversion efficiency (PCE) of 26.0% for single-junction devices and 33.7% for perovskite-based tandem solar cells (NREL Best Research-Cell Efficiency Chart).

A significant fraction of the record efficiency single junction PSCs has relied on the original n–i–p device architecture. However, there has been a steady growth in interest in the inverted p–i–n device

W. H. K. Perera, M. G. Masteghin, J. S. Yun, W. Zhang, K. D. G. I. Jayawardena, S. R. P. Silva
Advanced Technology Institute
School of Computer Science and Electronic Engineering
University of Surrey
Guildford, Surrey GU2 7XH, UK
E-mail: s.silva@surrey.ac.uk

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/solr.202300388.

DOI: 10.1002/solr.202300388

H. Shim
Australian Centre for Advanced Photovoltaics
School of Photovoltaic and Renewable Energy Engineering
University of New South Wales
Sydney, NSW 2052, Australia

J. D. Davies, J. L. Ryan
Department of Chemistry
University of Warwick
Coventry CV4 7AL, UK

S. J. Hinder
The Surface Analysis Laboratory
Faculty of Engineering and Physical Sciences
University of Surrey
Guildford, Surrey GU2 7XH, UK

architecture. This has been driven by several factors including the low processing temperatures generally used for the p–i–n architecture, which ensures its compatibility with roll-to-roll processing as well as its relevance for tandem architectures which has recently led to PCEs that exceed the single-junction radiative limit for silicon PVs.[4,5]

The development of inverted device architectures has mainly relied on the use of organic hole-transport layers (HTLs) such as poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine (PTAA) and poly[(*N*,*N*′-bis-(4-butylphenyl)-*N*,*N*′-bis(phenyl)benzidine] (poly-TPD).[6–8] These organic HTLs have resulted in lower recombination losses in PSCs allowing high open-circuit voltages (V_{OC}) to be realized in inverted devices.[7,9] Recently, Albrecht and co-workers developed a class of self-assembled monolayers (SAMs) that eliminates the requirement of conventionally used transport layers at the anode contact in PSCs.[10] Among these, the SAMs 2-(3,6-dimethoxy-9*H*-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) and (2-(9*H*-carbazol-9-yl)ethyl)phosphonic acid (2PACz) have been widely adopted in place of conventional organic HTLs, yielding promising results in perovskite-based single-junction and tandem devices.[11–17] This was followed by the development of the methyl-substituted SAM, [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) (**Figure 1**a).[11] Compared to the previously developed SAMs, Me-4PACz has its highest occupied molecular orbital (HOMO) more favorably positioned with respect to the valence band maximum (VBM) of the perovskite. This leads to a more efficient hole extraction at the hole selective contact (HSC)/perovskite interface compared to MeO-2PACz, 2PACz and even PTAA, which translated to lower V_{OC} losses and improved resistance to light-induced phase segregation.[11]

A key challenge associated with the use of organic HTLs in PSCs has been the poor wettability of the perovskite precursor which leads to the formation of perovskite films with pinholes. This has, in the past, been addressed by solvent treating with dimethylformamide or by using poly[(9,9-bis(3′-((*N*,*N*-dimethyl)-*N*-ethylammonium)-propyl)-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene)]dibromide (PFN-Br, Figure 1b) as an interfacial compatibilizer.[18–20] Another example of an interfacial compatibilizer used with hydrophobic charge transport layers is alumina (Al_2O_3) nanoparticles (NPs), although this has not been widely adopted.[21–25] Similar challenges associated with the wettability of the perovskite have been reported for coating of perovskite precursors on Me-4PACz by several groups which have also been observed by us.[26–28] This has led to a lower number of reported studies based on the use of Me-4PACz on PSCs due to poor device reproducibility. One route toward improving the device yield on Me-4PACz is texturing of the underlying substrate as reported by Tockhorn et al.[28] While this approach is compatible with tandem architectures, there is a need for

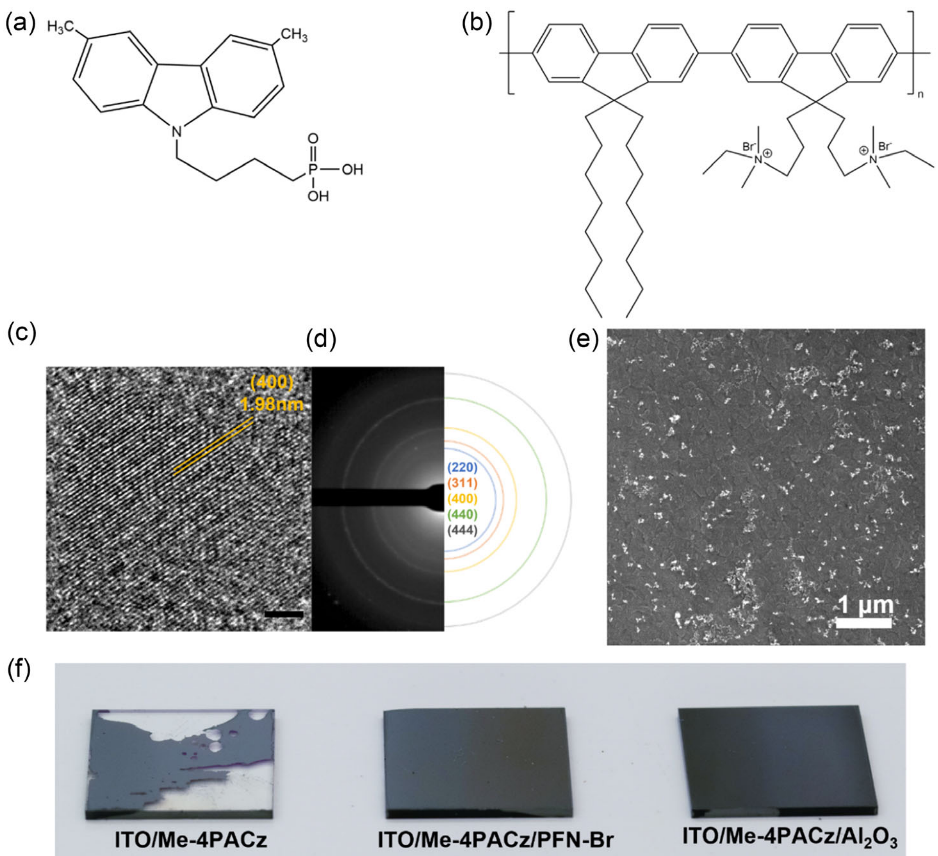

Figure 1. Molecular structure of a) Me-4PACz and b) PFN-Br. c) A high-resolution TEM and d) the related SAED pattern of the Al_2O_3 NPs used in this study. e) Scanning electron micrograph indicating the surface coverage of Al_2O_3 NPs on Me-4PACz-coated ITO. f) Photographs showing the poor wettability of perovskite on bare Me-4PACz which is significantly improved upon modification with PFN-Br and Al_2O_3.

methodologies that can be implemented on single-junction PSCs.

Here, we introduce interfacial modification with randomly dispersed, solution processable Al_2O_3 NPs as means of overcoming the wettability challenge of perovskite on the hydrophobic Me-4PACz over large areas (up to 4 cm^2 demonstrated). We show that the Al_2O_3 NPs act as pinning sites for the perovskite precursor resulting in complete coverage of the perovskite absorber on the SAM. Furthermore, we show that modification with Al_2O_3 NPs results in perovskites with Shockley–Read–Hall (SRH) recombination lifetimes exceeding 3 μs, which is higher than those on perovskites coated on bare Me-4PACz and on PFN-Br-modified Me-4PACz. We propose that this phenomenon is in accordance with the amphoteric nature of Al_2O_3 which allows it to behave as both a Lewis acid and a Lewis base. PSCs fabricated on Me-4PACz modified with Al_2O_3 NPs demonstrate a champion PCE of 19.9% which is a marked improvement in PCE as compared to 16.5% observed on devices with PFN-Br-modified Me-4PACz. We note that following the first submission of this manuscript, Al-Ashouri et al. have reported the addition of 1,6-hexylenediphosphonic acid as a second component into the Me-4PACz solution resulting in similar PCE for triple-cation perovskites reported herein.[29] Furthermore, Peng et al. have also reported the use of Al_2O_3 NPs where reduced recombination was demonstrated resulting in a PCE of ≈25% when combined with a perovskite with a narrower bandgap of 1.55 eV.[30]

2. Results and Discussion

2.1. Optimization of Al_2O_3 NP Coverage

We initially characterized the Al_2O_3 NP dispersion to identify its structure and size distribution. A high-resolution transmission electron microscopy (TEM) image of a representative Al_2O_3 NP is given in Figure 1c while the related selected area electron diffraction (SAED) pattern is given in Figure 1d in which the crystal planes have been indexed based on the γ-Al_2O_3 phase.[31] Using dynamic light scattering measurements, we identify the average particle size to be 17.1 ± 3.9 nm (Figure S1, Supporting Information). We then studied the impact of the NP concentration on the NP coverage on Me-4PACz (Figure S2, Supporting Information) and its effect on the PSC performance. While Lee et al. demonstrated that a mesoporous layer of Al_2O_3 NPs results in a faster charge extraction compared to titania in regular PSCs,[32] the work of Stranks et al. demonstrated that the presence of Al_2O_3 NPs can result in suboptimal performance.[33] From our initial optimization studies, we identified a 1:1000 dilution of the parent Al_2O_3 NP dispersion as the optimum concentration for realizing a balance between improving device reproducibility and maintaining device efficiency (Figure 1e, S3, S4, Table S1, Supporting Information). Compact (i.e., pinhole free) perovskite films were achieved over large areas (demonstrated up to 4 cm^2), both when processed on Me-4PACz modified under this optimized Al_2O_3 condition as well as on Me-4PACz modified with PFN-Br, whereas the absence of a surface modifier resulted in incomplete coverage (Figure 1f) of the perovskite absorber leading to a high number of shunted devices.

2.2. Precursor Retention and Surface Chemistry

To understand the impact of the different surface modifiers on the retention of the perovskite precursor on Me-4PACz, we carried out roll-off angle measurements, as conducted by Tockhorn et al.[28] In our measurements, we observed a roll-off angle of 30.9° for Me-4PACz on indium tin oxide (ITO) which is significantly improved to 37.2° when Me-4PACz is modified with PFN-Br. On the other hand, modification of Me-4PACz with Al_2O_3 leads to a roll-off angle of only 31.9° (**Figure 2**a). Although this roll-off angle is not as high as that for PFN-Br, the low density of randomly positioned Al_2O_3 NPs on the substrate surface is sufficient to pin the precursor on the SAM, thereby allowing a combination of high device yield and excellent optoelectronic properties. We further observe breaking up of the perovskite precursor into smaller droplets on bare Me-4PACz while the precursor is observed to maintain its cohesiveness and show better wettability on Me-4PACz modified with PFN-Br and Al_2O_3 (Figure 2b). The enhanced retention can be explained based on the mathematical model developed by Joanny and de Gennes which describes the influence of randomly dispersed particles (or contaminants) on the surface.[34,35] However, we do note that perovskite precursors with their multicomponent nature, in combination with the complex surface chemistry of the underlying substrate, will require the development of more detailed models.

Following the above, we proceeded to understand the surface chemistry of the modified substrates using X-ray photoelectron spectroscopy (XPS) measurements. In the P 2*p* and N 1*s* spectra given in Figure S5, Supporting Information, the peaks appearing at binding energies of 133.9 and 400 eV correspond to the P in phosphate group and N in C—N bonds, respectively, of Me-4PACz,[36] confirming the presence of the SAM on the ITO. Further, In $3d_{3/2}$ and In $3d_{5/2}$ signals characteristic to ITO at binding energies of ≈452 and ≈445 eV, respectively, can be observed in the In 3*d* spectra of the Me-4PACz-coated substrates (Figure S6a, Supporting Information).[37] A probing depth of ≈5 nm was used in these measurements. Based on the appearance of peaks specific to ITO in SAM-coated substrates at such small probing depths, it is reasonable to assume that a few nm-thick layer of Me-4PACz, i.e., a monolayer, is formed here. The O 1*s* and In 3*d* spectra depicted in Figure 2c and S6a, Supporting Information, respectively, show a slight, but noticeable shift in the signals toward higher binding energies in substrates coated with Me-4PACz (in comparison to bare ITO). This is consistent with observations reported in the literature where it has been correlated to a change in the work function due to the introduction of a dipole by the SAM.[37]

To evaluate the effect of the Me-4PACz modification on surface roughness and work function (Φ) of ITO, we carried out atomic force microscopy (AFM) and scanning Kelvin probe force microscopy (SKPFM). As shown in Figure 2d and **Table 1**, Me-4PACz causes the root mean square roughness (R_q) of ITO to increase from ≈3.6 to ≈6.6 nm while modification with Al_2O_3

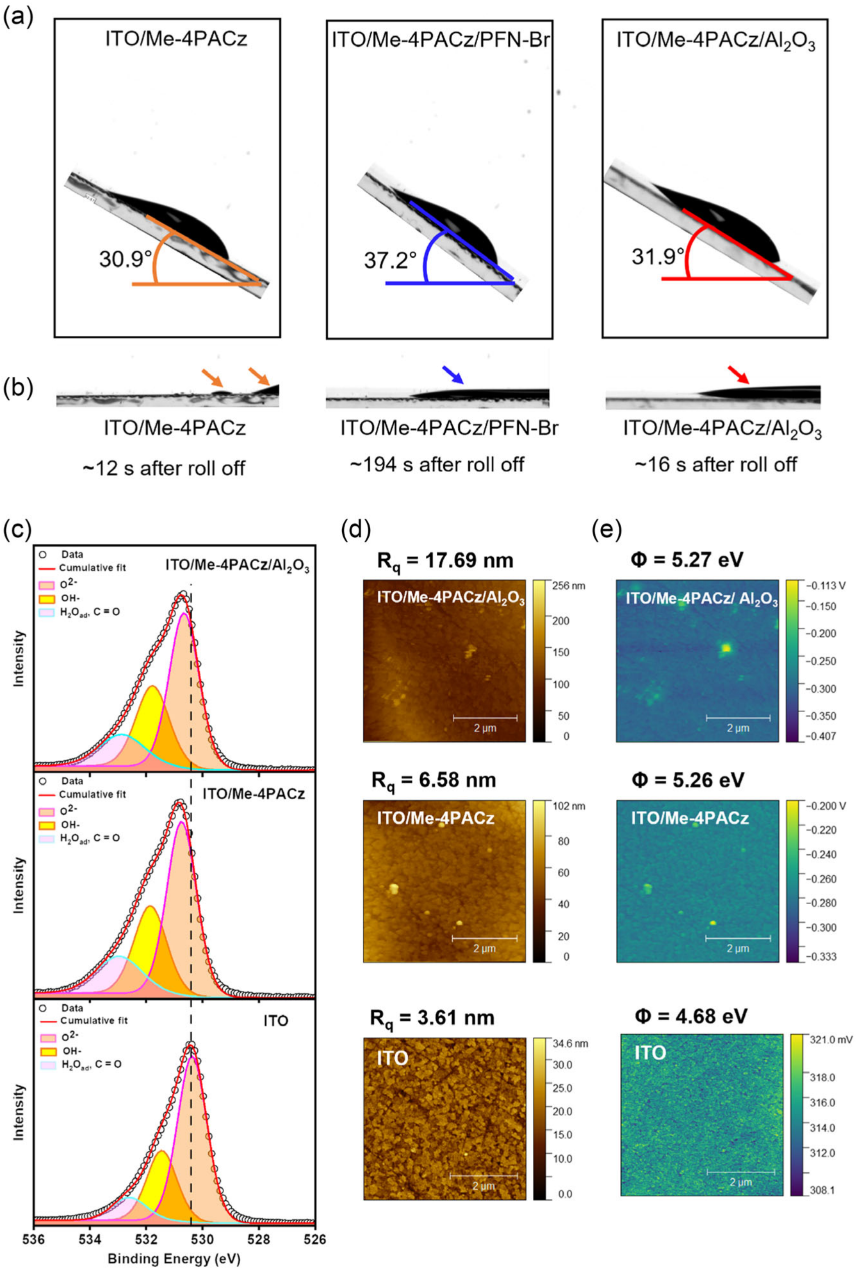

Figure 2. a) Roll-off angle measurements of $Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$ on ITO/Me-4PACz, ITO/Me-4PACz/PFN-Br, and ITO/Me-4PACz/Al_2O_3. The photographs have been captured at the point at which roll-off is initiated. The angles are indicated as a guide to the eye. b) Photographs from roll-off angle measurements indicative of the dewetting of the precursor on Me-4PACz and improved wetting on Me-4PACz modified with PFN-Br and Al_2O_3 NPs. c) O 1s XPS spectra, d) AFM height images, and e) CPD mapping of ITO, ITO/Me-4PACz and ITO/Me-4PACz/Al_2O_3.

Table 1. Root mean square roughness and work function values of ITO, ITO/Me-4PACz, and ITO/Me-4PACz modified with PFN-Br and Al_2O_3 NPs.

Sample	R_q [nm]	Work function [eV]
ITO	3.6	4.68
ITO/Me-4PACz	6.6	5.26
ITO/Me-4PACz/PFN-Br	5.7	4.89
ITO/Me-4PACz/Al_2O_3 NP	17.7	5.27

causes R_q to further increase to ≈17.7 nm. The latter indicates the presence of Al_2O_3 NPs on Me-4PACz. Figure 2e shows the corresponding contact potential difference (CPD) mapping of the samples, with the Φ values given in Table 1. It is apparent that, while the work function of ITO increases from ≈4.7 to ≈5.3 eV when coated with Me-4PACz, it is not altered further upon modification with Al_2O_3. Furthermore, it can be inferred from the homogeneity of the CPD map that there is uniform and homogeneous spatial distribution of Me-4PACz on the substrates, which is favorable toward achieving efficient charge extraction.[38]

2.3. Surface and Bulk Structure

To gain an insight into the influence of the Me-4PACz surface modification on the crystallization and the morphology of the perovskite, we studied the films using scanning electron microscopy (SEM) and grazing incidence X-ray diffractometry (GIXRD). Analysis of the scanning electron micrographs (**Figure 3**a–c) for grain sizes (Figure 3d–f) shows that perovskite grains exceeding 240 nm in diameter are formed on modified Me-4PACz as compared to 196 nm on bare Me-4PACz. The formation of large grains is considered as favorable for device performance as it reduces the trap density at grain boundaries which act as non-radiative recombination sites for charges carriers.[39] However, we note that these grain sizes may not extend through the film thickness and can vary depending on the nature of the substrate leading to differences in carrier lifetimes as observed in time-resolved photoluminescence (TRPL) measurements which will be discussed later. GIXRD patterns of perovskites formed directly on Me-4PACz (as a reference) and on modified Me-4PACz are depicted in Figure 3g. One of the most notable differences between the spectra is the peak at 30.3° which only appears in the spectra of ITO/Me-4PACz. This peak, which is commonly attributed to ITO,[40] confirms the presence of pinholes/areas not covered by the perovskite when coated directly on Me-4PACz. The absence of such an intense peak for ITO in the spectra for modified Me-4PACz substrates indicates the achievement of an improved coverage of the perovskite by modification using both PFN-Br and Al_2O_3. It is further noted that the peak at 12.7° corresponding to PbI_2 appears more prominently on ITO/Me-4PACz/PFN-Br and ITO/Me-4PACz/Al_2O_3 as compared to that

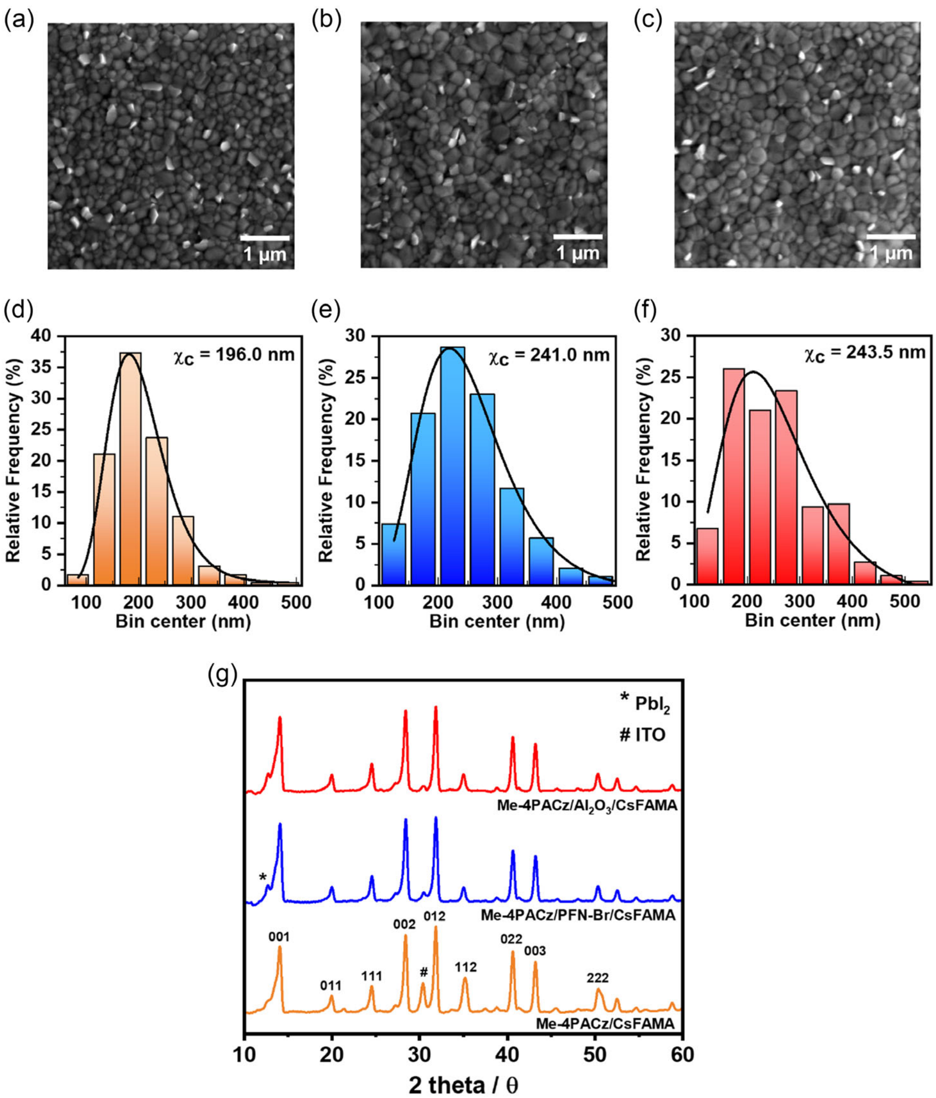

Figure 3. SEM images of perovskite films formed on a) Me-4PACz, b) Me-4PACz/PFN-Br, and c) Me-4PACz/Al_2O_3. Perovskite grain size analysis on d) Me-4PACz, e) Me-4PACz/PFN-Br, and f) Me-4PACz/Al_2O_3. χ_c = mean bin center. g) GIXRD patterns of perovskite films formed on Me-4PACz and on Me-4PACz modified with PFN-Br and Al_2O_3.

on ITO/Me-4PACz.[41] This indicates the presence of excess PbI_2 in perovskite films formed on modified Me-4PACz. Excess PbI_2 is often incorporated in high-performing PSCs as it is reported to have beneficial effects on grain boundary trap passivation which is also expected to be a contributing factor toward the realization of longer SRH recombination lifetimes in perovskites formed on Me-4PACz/Al_2O_3 as compared to those on bare Me-4PACz.[42]

2.4. Device Performance

To examine the effect of this Al_2O_3-based HSC/perovskite interfacial modification on device performance, we fabricated inverted planar PSCs with a configuration of ITO/Me-4PACz/$Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$/C60/bathocuproine (BCP)/Ag (**Figure 4**a). It should be noted here that, as discussed initially, the dewetting of perovskite on Me-4PACz was so severe that it was not possible to achieve working devices on Me-4PACz without any modification. Figure 4b depicts the current density (J)–voltage (V) characteristics of champion devices on Me-4PACz modified with PFN-Br and Al_2O_3 under AM1.5G illumination with a summary of the related photovoltaic parameters in **Table 2**. The champion device fabricated on Me-4PACz modified with Al_2O_3 NPs showed a significantly improved performance with a PCE of 19.9% as compared to the one fabricated on PFN-Br (16.5%). The corresponding external quantum efficiency (EQE) curves of these devices are given in Figure 4c where an improved EQE is observed for the Al_2O_3 device in the 400–700 nm wavelength range which explains its higher short circuit photocurrent density (J_{SC}, 23.0 mA cm^{-2}) compared to that of the PFN-Br device (20.5 mA cm^{-2}). This higher J_{SC} is in agreement with the improved charge carrier lifetimes observed on Al_2O_3-modified Me-4PACz in comparison to those on PFN-Br-modified Me-4PACz in TRPL measurements shown in **Figure 5**a. We also tracked any changes in the current density of the champion devices at their maximum power point (MPP) for 180 s (Figure 4d). Here, a satisfactorily stable current was observed in both devices during the testing period indicative of sufficient device stability. The statistical distribution of the photovoltaic parameters for Al_2O_3 devices ($N = 33$) is given in Figure 4e–h with their average values summarized in Table S2, Supporting information (the corresponding statistical distributions for PFN-Br-modified devices are given in Figure S7, Supporting Information). Across all device parameters, Al_2O_3-modified devices show significantly higher average performances as compared to PFN-Br-modified devices. Figure 4i,j depicts the reflectance and EQE spectra as well as photocurrent density losses due to reflectance and electronic effects in champion devices fabricated on Me-4PACz modified with PFN-Br and Al_2O_3, respectively. Here, while there seem to be similar losses due to reflectance in the two devices, electronic losses are noticeably lower in the Al_2O_3 device (≈2.0 mA cm^{-2}) as compared to the PFN-Br device (≈2.7 mA cm^{-2}). The higher losses in the PFN-Br-based devices are attributed to the higher nonradiative losses as evidenced through optical and electrical characterization which will be discussed later. We note a similar trend for $MAPbI_3$-based PSCs when fabricated on Al_2O_3 NP-modified Me-4PACz as compared to devices fabricated on PFN-Br-modified Me-4PACz (Figure S8–S10, and Table S3, Supporting Information).

2.5. Optoelectronic Origins of the Observed Performance Improvements

To identify any influence of the surface treatment of Me-4PACz on the recombination characteristics, we conducted both steady-state PL and TRPL measurements on perovskite films coated on Me-4PACz, Me-4PACz/PFN-Br, and Me-4PACz/Al_2O_3. TRPL measurements showed a shorter SRH recombination lifetime of ≈1 μs for perovskites coated on Me-4PACz/PFN-Br (Figure 5a and Table S4, Supporting Information) in comparison to films coated directly on Me-4PACz which showed longer recombination lifetimes of ≈1.4 μs. More importantly, SRH recombination lifetimes exceeding 3.3 μs were observed in perovskites formed on Me-4PACz/Al_2O_3 (under optimized Al_2O_3 coverage), indicating further suppression of SRH recombination (compared to films coated on unmodified Me-4PACz).

Within the field of perovskite optoelectronics, Lewis acids and Lewis bases are well known and widely adopted for passivation of defects in the perovskite absorber.[43] Al_2O_3, due to its amphoteric nature, can behave as both a Lewis acid and a Lewis base which enables it to cause defect passivation of perovskites. A typical reaction scheme which allows this to happen is given in Figure 5b(i).[44] During thermal treatment of Al_2O_3, any hydroxy (OH) groups bonded to the surface are detached. When this occurs at two neighboring sites, a strained oxygen bridge is formed resulting in a local Lewis acid and Lewis base behavior as shown.[45] In the examples depicted in Figure 5b(ii) and (iii), Pb–I antisite and halide defects are passivated by Lewis acids due to their ability to accept electrons from negatively charged sites while defects such as halide vacancies or unsaturated Pb^{2+} defects are passivated by Lewis bases due to their ability to donate electrons to defect sites.[46] Based on the above, we propose that the observed improvements in the SRH recombination lifetimes for perovskites coated on Me-4PACz are due to the passivation of defects at HSC/perovskite interface, facilitated by the amphoteric nature of Al_2O_3. We note that the nature of the substrate, i.e., Me-4PACz, Me-4PACz/PFN-Br, or Me-4PACz/Al_2O_3, does not affect the bandgap of the perovskite absorber as is evident through the steady-state PL spectra of the films depicted in Figure 5c where no noticeable differences in the peak position or the shape can be observed. Further, the bandgap estimated here is in agreement with the values reported in the literature for this composition.[41]

Following the characterization and analysis of the PL characteristics, the V_{OC} behavior of the devices with varying illumination intensity was examined to understand the influence of the hole extraction contact modification on the electronic characteristics of the devices. The diode ideality factors for PSCs based on Me-4PACz/PFN-Br and Me-4PACz/Al_2O_3 were estimated using the following relationship[47]

$$n = \frac{q}{kT}\frac{d(V_{OC})}{d(\ln\phi)} \tag{1}$$

Here, n, k, q, T, and ϕ represent the diode (light) ideality factor, Boltzmann constant, electronic charge, temperature, and light intensity, respectively. Based on a linear fit to a semilogarithmic plot of V_{OC} versus $\log(\phi)$ (Figure 5d), we estimate a lower n value of ≈1.38 for devices based on Me-4PACz/Al_2O_3 as opposed to a

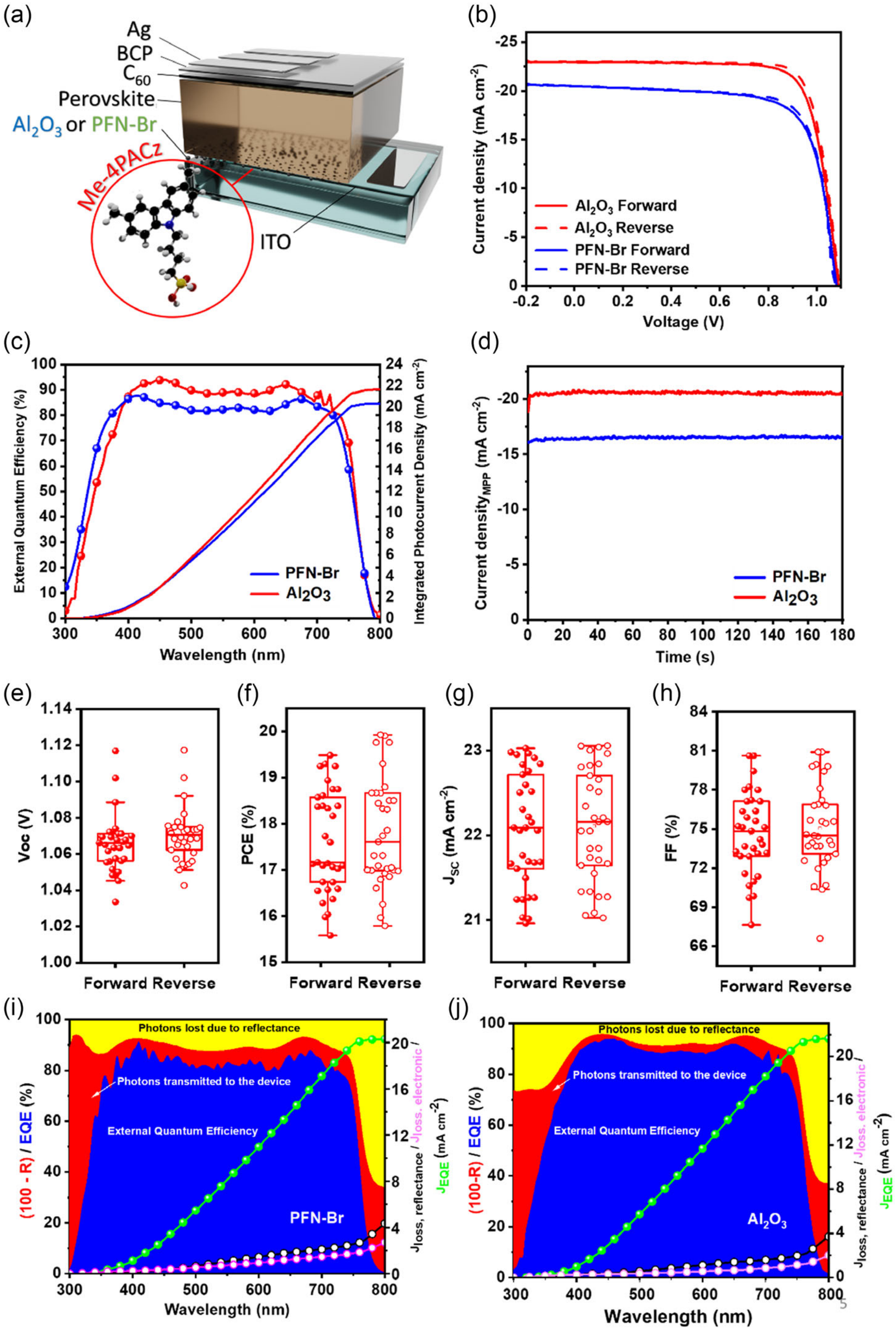

Figure 4. a) Device architecture of the PSCs. b) J–V curves of the champion devices fabricated on PFN-Br- and Al_2O_3-modified Me-4PACz under AM1.5G illumination. c) Corresponding EQE spectra and integrated photocurrent densities (J_{EQE}). d) Stability testing of champion devices by tracking their photocurrent densities (J_{SC}'s) at MPP for 180 s. e–h) Statistical distribution of photovoltaic parameters for Al_2O_3 devices. EQE, J_{EQE}, reflectance (R), and electronic losses of representative devices fabricated on i) PFN-Br- and j) Al_2O_3-based modification. $J_{loss,\ reflectance} = J_{SC}$ losses due to reflectance and $J_{loss,\ electronic} = J_{SC}$ losses due to electronic effects.

higher n value of ≈1.70 for devices when Me-4PACz is modified using PFN-Br, indicative of more recombination within the device for the latter architecture. The trend for n value estimated here agrees with the longer SRH lifetimes observed for perovskite films on Me-4PACz modified with Al_2O_3 in TRPL measurements. Furthermore, the light intensity dependence of J_{SC} was analyzed for both device types. Based on a linear fit for log (J_{SC}) versus log(ϕ), we obtained a slope of 0.95 and 0.94 for Me-4PACz modified with Al_2O_3 and PFN-Br, respectively (Figure S11, Supporting Information), indicative of low

Table 2. Photovoltaic parameters of champion $Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$ devices fabricated on PFN-Br- and Al_2O_3-based modification of Me-4PACz.

Modification	Scan direction	V_{OC} [V]	J_{SC} [mA cm^{-2}]	FF [%]	PCE [%]
PFN-Br	Forward	1.08	20.5	72.5	16.0
	Reverse	1.08	20.5	74.7	16.5
Al_2O_3	Forward	1.09	23.0	77.2	19.3
	Reverse	1.09	23.0	79.4	19.9

bimolecular recombination under short-circuit conditions. We also analyzed the electrical characteristics of hole-only devices based on ITO/Me-4PACz/PFN-Br or Al_2O_3/perovskite/Spiro-OMeTAD/Au. Plotting $\log(J) \approx \log(V)$, we observe three different regions: an Ohmic region characterized by a slope of 1 (or $J \propto V$) followed by a rapid rise indicative of a $J \propto V^m$ type behavior ($m > 2$) and thirdly, the emergence of a quadratic behavior ($J \propto V^2$) indicative of the Child's regime for both device types (Figure 5e,f). We observe a reduction in the voltage for the trap-filled limit (V_{TFL}) when Me-4PACz is modified with Al_2O_3

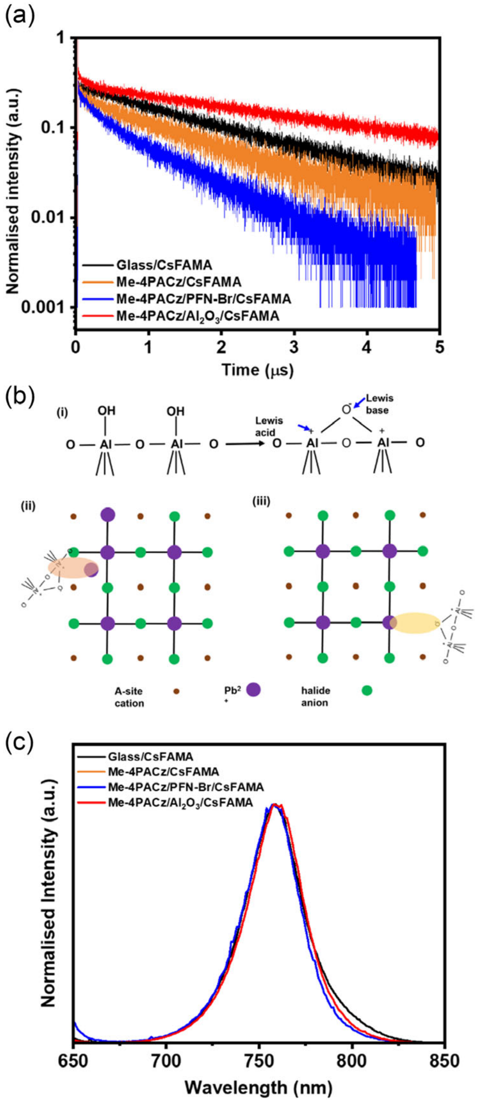

Figure 5. a) TRPL measurements of perovskite films on glass, Me-4PACz, Me-4PACz/PFN-Br, and Me-4PACz/Al_2O_3. b) (i) Origin of Lewis acid and Lewis base behavior in Al_2O_3. Examples of passivation of (ii) an undercoordinated Pb defect and (iii) a halide vacancy by Al_2O_3 through its Lewis acid and Lewis base behavior. c) Normalized steady-state PL spectra of perovskite films on glass, Me-4PACz, Me-4PACz/PFN-Br, and Me-4PACz/Al_2O_3 (the orange line representing the PL intensity of Me-4PACz/$Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$ is not visible in the figure due to overlapping with the red line representing the PL intensity of Me-4PACz/Al_2O_3/$Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$). No significant shift in the bandgap of the perovskite is observed. d) V_{OC} versus light intensity plot for PSCs fabricated on Me-4PACz/PFN-Br and Me-4PACz/Al_2O_3. The diode ideality factors calculated from a linear fit are indicated. Current density versus voltage curves for hole-only devices fabricated on e) Me-4PACz/Al_2O_3 and f) Me-4PACz/PFN-Br. The black, green, and yellow lines indicate linear fits to the Ohmic, $J \propto V^m$ where $m > 3$ and Child regime. The V_{TFL} values as estimated from the intersection point of the former two regions are also indicated.

NPs compared to modification with PFN-Br. We estimated the trap density (N_t) for the two systems using the following relationship[48]

$$N_t = \frac{2\varepsilon\varepsilon_o V_{TFL}}{eL^2} \quad (2)$$

Here, ε, ε_o, and L represent the relative permittivity (taken as 62),[48] vacuum permittivity, and thickness of the perovskite absorber (≈500 nm, Table S5, Supporting Information), respectively. We note here that we did not observe a change in the perovskite thickness irrespective of whether PFN-Br or Al_2O_3 NPs were used to modify the Me-4PACz surface. Based on the above values, we estimate trap densities of $\approx 5.7 \times 10^{15}$ and $\approx 3.3 \times 10^{15}$ cm^{-3} when Me-4PACz is modified with PFN-Br and Al_2O_3 NPs, respectively, in agreement with our previous observations in TRPL and V_{OC} versus log(ϕ) measurements. This reduction in trap density for perovskites on Me-4PACz modified with Al_2O_3 NPs in comparison to perovskites on Me-4PACz modified with PFN-Br leads to the higher J_{SC} observed for the former type of devices (as evident in Figure 5).

Finally, we evaluated the influence of the Me-4PACz surface modification on the stability of these devices under ISOS-D-2 and ISOS-D-2-i conditions (i.e., stored under open-circuit conditions in the dark at a temperature of 65 °C in ambient and N_2 environment, respectively).[49] Within the duration studied, the PFN-Br-modified devices show a higher degradation in device performance in comparison to Al_2O_3 NP-modified devices (Figure S12, Supporting Information). In particular, the PFN-Br-modified devices demonstrate more than 20% drop from the initial PCE, while the Al_2O_3 NP-modified devices retained more than 90% of the initial PCE under ambient conditions. These preliminary results indicate the promise of Al_2O_3 NP modification as a strategy toward realizing stable inverted PSCs based on hydrophobic SAMs.

3. Conclusion

In conclusion, we have demonstrated that the incorporation of Al_2O_3 NPs at the HSC/perovskite interface is an effective strategy toward achieving better, uniform coverage of perovskites over large areas in Me-4PACz-based inverted PSCs. In comparison to other more commonly used interfacial compatibilizers such as PFN-Br, the use of Al_2O_3 NPs results in the realization of significantly longer carrier lifetimes, even exceeding those of perovskites coated directly on Me-4PACz, mediated by a reduction in trap density. This, in turn, leads to higher device efficiencies approaching 20%. The use of such insulating NPs as pinning agents for the perovskite precursor is anticipated to not only be beneficial on Me-4PACz, but also on other organic transport layers such as PTAA and poly-TPD allowing the realization of higher efficiencies on many different optoelectronic systems that use perovskites as its active layer.

4. Experimental Section

Materials: Formamidinium iodide (FAI, ≥99.99%), methylammonium bromide (MABr, ≥99.99%), and methylammonium iodide (MAI, >99.99%) were purchased from GreatCell Solar Materials, Australia. Cesium iodide (CsI, 99.999%) and aluminum oxide (Al_2O_3, 20% v/v in 2-propanol) were purchased from Sigma–Aldrich (UK). Lead(II) iodide (PbI2, 99.99%) and Lead(II) bromide (PbBr2, >98.0%) were purchased from Tokyo Chemical Industry Co., Ltd. (TCI, Japan). Poly[(9,9-bis(3'-((*N*,*N*-dimethyl)-*N*-ethylammonium)-propyl)-2,7-fluorene)-alt-2,7-(9,9-dioctylfluorene)]dibromide (PFN-Br) was purchased from 1-material, Canada. [4-(3,6-dimethyl-9*H*-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) was purchased from Dyenamo, Sweden. C60 and bathocuproine (BCP, sublimed grade >99.5%) were purchased from Ossila, UK. Solvents dimethyl sulfoxide (DMSO, anhydrous, ≥99.9%), *N*,*N*-dimethylformamide (DMF, anhydrous, 99.8%), chlorobenzene (anhydrous, 99.8%), methanol (HPLC grade, ≥99.9%), ethanol (anhydrous, >99.5%) and 2-propanol (anhydrous, 99.5%) were purchased from Sigma–Aldrich, UK.

Substrate Cleaning and Preparation: ITO-coated glass substrates purchased from South China Science and Technology Ltd. (20 mm × 20 mm with a thickness of 1.1 mm and a sheet resistance of 15 Ω □$^{-1}$) were first cleaned by sonicating in a 2% v/v Hellmanex in water solution for 30 min. The substrates were then rinsed with deionized water and sonicated in water for further 30 min. Thereafter, they were sequentially cleaned in acetone, 2-propanol, and methanol in an ultrasonic bath at ≈40 °C for 15 min each and blow dried with N2. Prior to coating with Me-4PACz, the substrates were subjected to a UV–O_3 pretreatment for 10 min using a Jelight UV–O_3 Cleaner Model 24.

Me-4PACz: A 1 mmol L^{-1} solution of Me-4PACz was prepared in ethanol and left to stir overnight inside the glove box.

PFN-Br: A 0.5 mg mL^{-1} solution of PFN-Br was prepared in methanol and left to stir overnight inside the glove box.

Al_2O_3: Al_2O_3 20% v/v in 2-propanol, Sigma–Aldrich was diluted as required in 2-propanol.

$MAPbI_3$: PbI_2 (461 mg) and MAI (159 mg) were weighed into one vial. To this, DMF (595 μL) and DMSO (71 μL) were added and left to stir overnight in the glove box at room temperature. The precursor was allowed to stir at 65 °C for 15 min prior to deposition and maintained at 65 °C during deposition.

$Cs_{0.05}MA_{0.15}FA_{0.8}Pb(I_{0.85}Br_{0.15})_3$: PbI_2 (479.5 mg, 1.04 M), FAI (165 mg, 0.96 M), $PbBr_2$ (67.9 mg, 0.185 M), MABr (20.2 mg, 0.18 M), and CsI (15.6 mg, 0.06 M) were weighed into one vial. To this, DMF:DMSO (1 mL) at a 4:1 volume ratio was added. The precursor was left stirring overnight in the glove box at room temperature.

Device Fabrication: Spin coating of Me-4PACz on the substrates was carried out immediately following UV–O_3 treatment to ensure adhesion of the SAM to the substrate. 50 μL of the solution was spread on the substrate and after a waiting time of 5 s, it was spun at 3000 rpm for 30 s. The films were then annealed at 100 °C for 10 min. For PFN-Br-based modification, 30 μL of the PFN-Br solution was dispensed on a substrate spinning at 5000 rpm, at 5 s from the spin start time while the overall spin coating duration was kept at 30 s. For Al_2O_3-based modification, 40 μL of the solution was deposited by spin coating at 4000 rpm for 30 s. The spin coating of perovskite was carried out immediately following the surface modification of Me-4PACz using PFN-Br or Al_2O_3. For $MAPbI_3$-coated substrates, 35 μL of the precursor was spread on the substrate followed by spinning at 4000 rpm for 30 s. 600 μL of diethylether, the antisolvent, was dropped at 7 s from the spin start time. CsFAMA perovskite precursor was spin-coated using an antisolvent-assisted method and a two-step spin program. 50 μL of the perovskite precursor was spread over the substrate prior to starting the spinning process. The substrates were first spun at 2000 rpm for 10 s and then at 3500 rpm for 30 s. At 10 s from the end of the spin program, 120 μL of CB was dispensed on the substrate as the antisolvent. The substrates were then transferred to a hot plate at 100 °C and annealed for 30 min. C60 (20 nm), BCP (7 nm) and Ag (100 nm) were thermally evaporated in an Angstrom EvoVac system. For devices fabricated for stability testing, Cu (100 nm) was used in place of Ag. For deposition of Cu, the samples deposited with C60 and BCP were taken out of the glove box and loaded to an evaporator (Moorfield) placed outside the glove box.

Current (I)–Voltage (V) Characteristics: *I–V* characteristics of the fabricated solar cells were evaluated using an Enlitech SS-F5-3A (Class 3A) solar simulator with a Keysight 2901A source measure unit acting as the electrical load. The calibration of the simulator was carried out using a KG-5 filtered Si diode. A mask with 0.09 cm^2 aperture area was used to define the active area of the device. All devices were measured without any encapsulation under ambient conditions at a temperature of ≈25 °C, light intensity of 100 mW cm^{-2} (AM1.5G), and a relative humidity of 30–35%. No preconditioning of the cells was carried out.

External and Internal Quantum Efficiency (EQE, IQE): EQE measurements of the fabricated devices were carried out using a Bentham PVE300 system. All measurements were carried out under ambient conditions. Reflectance measurements were carried out using an integrating sphere incorporated to the same system. All devices were measured without any encapsulation under ambient conditions at a temperature of ≈25 °C and a relative humidity of 30–35%.

Stability Testing: For stability testing, devices were encapsulated with a proprietary encapsulant developed in-house. For ISOS-D-2 testing, samples were stored at 65 °C in the dark under ambient conditions with a relative humidity of ≈35%. For ISOS-D-2-i testing, samples were stored at 65 °C in the dark in a N_2 glove box.

SEM: SEM images of the perovskite films were obtained using a TESCAN FERA3 dual beam/focused ion beam SEM under an accelerating voltage of 5 kV. Grain size analysis was carried out using ImageJ software.

GIXRD: GIXRD measurements were taken using a Panalytical X'pert Pro diffractometer using a GI stage with an incident angle of 1° using a Cu Kα_1 X-ray source driven at 45 kV.

XPS: XPS spectra were obtained on a Thermo Fisher Scientific Instruments K-Alpha+ spectrometer consisting of a monochromated Al Kα X-ray source ($h\nu = 1486.6$ eV) with a spot size of ≈400 μm radius. A pass energy of 200 eV was used for acquisition of the survey spectra. For obtaining high resolution core level spectra, a pass energy of 50 eV was used for all elements. For correction of possible charging effects that can occur during acquisition, the obtained spectra were charge referenced against the C 1s peak (285 eV). Fitting of the spectra was carried out using the manufacturers' Avantage software.

AFM: A commercial AFM system (Park Systems, NX10) was used for the KPFM measurements. Nitrogen gas was used for the ambient condition during the measurements and for cleaning the surface of the samples before the measurements. KPFM images were taken on the sample surface using a platinum-coated conductive probe (NSC15/Pt, force constant $k = 40$ N m^{-1}). For calibration, the work function of highly oriented pyrolytic graphite (HOPG) was measured using the same probe. During the measurements, DC bias was applied toward the tip. The KPFM images were taken with a scan direction from bottom to top and a scan rate of 0.8 Hz.

TEM and SAED: The Al_2O_3 NPs were diluted in 2-propanol and sonicated for 30 min. A 10 μL volume from the dispersed solution was dropped onto a Holey carbon support film and left to dry naturally. The structural analysis was carried out in a field emission transmission electron microscope (TEM) model Talos F200i, Thermo Scientific, using a 200 keV beam. Data analysis was done using the ImageJ software.

Steady-State and TRPL Spectroscopy: TRPL measurements were carried out using a Picoquant FT300 spectrophotometer. A 640 nm excitation source was used for both steady-state and TRPL measurements.

Roll-Off Angle Measurements: Roll-off angle (θ_{RA}) measurements were carried out using a Krüss DSA 100 drop shape analyzer equipped with tilt table. A single 10 μL droplet was deposited onto the test substrate (at a tilt angle of 0°) using the equipped dosing module and a Hamilton 1000 gastight syringe equipped with a 0.97 mm inner diameter dosing needle supplied by Weller. The substrate was tilted at a rate of 5° min^{-1} and the position of the deposited droplet was recorded as a function of time. θ_{RA} was determined for a given substrate-liquid system as the tilt angle corresponding to the initial movement of the three-phase point (3PP).

Thickness Measurements: Thickness measurements were carried out using a DektakXT profilometer.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

S.R.P.S. and W.Z. gratefully acknowledge support from the Engineering and Physical Sciences Research Council (EPSRC) Reference EP/V027131/1. S.R.P.S. acknowledges support through the CORNET program (Grant ID: 760949) from the European Union's Horizon 2020. S.R.P.S. and K.D.G.I.J. acknowledge support from the Equality Foundation of Hong Kong. W.H.K.P. gratefully acknowledges the University of Surrey Doctoral College for their support through the DCSA4 scheme. M.G.M. acknowledges support from the University of Surrey/Advanced Technology Institute via URS/ORS studentship. The authors acknowledge support from EPSRC for the use of Drop Shape Analyzer facility within the Warwick Analytical Service Centre (EP/V007688/1), University of Warwick, and the TEM facility at the University of Surrey (EP/V036327/1). K.D.G.I.J. acknowledges support from the University of Surrey for the CapEx scheme. The authors thank Prof. Robert Dorey from the School of Mechanical Engineering Sciences, University of Surrey for carrying out the Dynamic Light Scattering measurements and Thomas Webb from Imperial College London for assistance with the graphics.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

hydrophobicity, nanoparticles, perovskites, photovoltaics, self-assembled monolayers

Received: June 1, 2023
Revised: June 21, 2023
Published online: July 4, 2023

[1] A. S. R. Bati, Y. L. Zhong, P. L. Burn, M. K. Nazeeruddin, P. E. Shaw, M. Batmunkh, *Commun. Mater.* **2023**, *4*, 2.
[2] S. R. P. Silva, *Energy Environ. Mater.* **2021**, *4*, 497.
[3] J. P. Correa-Baena, M. Saliba, T. Buonassisi, M. Grätzel, A. Abate, W. Tress, A. Hagfeldt, *Science* **2017**, *358*, 739.
[4] D. Angmo, G. DeLuca, A. D. Scully, A. S. R. Chesman, A. Seeber, C. Zuo, D. Vak, U. Bach, M. Gao, *Cell Rep. Phys. Sci.* **2021**, *2*, 100293.
[5] J. Y. Hyun, K. M. Yeom, S. W. Lee, S. Bae, D. Choi, H. Song, D. Kang, J. K. Hwang, W. Lee, S. Lee, Y. Kang, H. S. Lee, J. H. Noh, D. Kim, *ACS Appl. Energy Mater.* **2021**, *5*, 5449.
[6] Y.-H. Lin, N. Sakai, P. Da, J. Wu, H. C. Sansom, A. J. Ramadan, S. Mahesh, J. Liu, R. D. J. Oliver, J. Lim, L. Aspitarte, K. Sharma, P. K. Madhu, A. B. Morales-Vilches, P. K. Nayak, S. Bai, F. Gao, C. R. M. Grovenor, M. B. Johnston, J. G. Labram, J. R. Durrant, J. M. Ball, B. Wenger, B. Stannowski, H. J. Snaith, *Science* **2020**, *369*, 96.
[7] D. Luo, W. Yang, Z. Wang, A. Sadhanala, Q. Hu, R. Su, R. Shivanna, G. F. Trindade, J. F. Watts, Z. Xu, T. Liu, K. Chen, F. Ye, P. Wu, L. Zhao, J. Wu, Y. Tu, Y. Zhang, X. Yang, W. Zhang, R. H. Friend, Q. Gong, H. J. Snaith, R. Zhu, *Science* **2018**, *360*, 1442.
[8] C. Bi, Q. Wang, Y. Shao, Y. Yuan, Z. Xiao, J. Huang, *Nat. Commun.* **2015**, *6*, 7747.
[9] M. Stolterfoht, P. Caprioglio, C. M. Wolff, J. A. Márquez, J. Nordmann, S. Zhang, D. Rothhardt, U. Hörmann, Y. Amir,

A. Redinger, L. Kegelmann, F. Zu, S. Albrecht, N. Koch, T. Kirchartz, M. Saliba, T. Unold, D. Neher, *Energy Environ. Sci.* **2019**, *12*, 2778.
[10] A. Magomedov, A. Al-Ashouri, E. Kasparavičius, S. Strazdaite, G. Niaura, M. Jošt, T. Malinauskas, S. Albrecht, V. Getautis, *Adv. Energy Mater.* **2018**, *8*, 1801892.
[11] A. Al-Ashouri, E. Köhnen, B. Li, A. Magomedov, H. Hempel, P. Caprioglio, J. A. Márquez, A. B. M. Vilches, E. Kasparavicius, J. A. Smith, N. Phung, D. Menzel, M. Grischek, L. Kegelmann, D. Skroblin, C. Gollwitzer, T. Malinauskas, M. Jošt, G. Matič, B. Rech, R. Schlatmann, M. Topič, L. Korte, A. Abate, B. Stannowski, D. Neher, M. Stolterfoht, T. Unold, V. Getautis, S. Albrecht, *Science* **2020**, *370*, 1300.
[12] A. Al-Ashouri, A. Magomedov, M. Roß, M. Jošt, M. Talaikis, G. Chistiakova, T. Bertram, J. A. Márquez, E. Köhnen, E. Kasparavičius, S. Levcenco, L. Gil-Escrig, C. J. Hages, R. Schlatmann, B. Rech, T. Malinauskas, T. Unold, C. A. Kaufmann, L. Korte, G. Niaura, V. Getautis, S. Albrecht, *Energy Environ. Sci.* **2019**, *12*, 3356.
[13] K. O. Brinkmann, T. Becker, F. Zimmermann, C. Kreusel, T. Gahlmann, M. Theisen, T. Haeger, S. Olthof, C. Tückmantel, M. Günster, T. Maschwitz, F. Göbelsmann, C. Koch, D. Hertel, P. Caprioglio, F. Peña-Camargo, L. Perdigón-Toro, A. Al-Ashouri, L. Merten, A. Hinderhofer, L. Gomell, S. Zhang, F. Schreiber, S. Albrecht, K. Meerholz, D. Neher, M. Stolterfoht, T. Riedl, *Nature* **2022**, *604*, 280.
[14] Q. Jiang, J. Tong, Y. Xian, R. A. Kerner, S. P. Dunfield, C. Xiao, R. A. Scheidt, D. Kuciauskas, X. Wang, M. P. Hautzinger, R. Tirawat, M. C. Beard, D. P. Fenning, J. J. Berry, B. W. Larson, Y. Yan, K. Zhu, *Nature* **2022**, *611*, 278.
[15] R. Azmi, E. Ugur, A. Seitkhan, F. Aljamaan, A. S. Subbiah, J. Liu, G. T. Harrison, M. I. Nugraha, M. K. Eswaran, M. Babics, Y. Chen, F. Xu, T. G. Allen, A. ur Rehman, C. L. Wang, T. D. Anthopoulos, U. Schwingenschlögl, M. De Bastiani, E. Aydin, S. De Wolf, *Science* **2022**, *376*, 73.
[16] M. A. Mahmud, J. Zheng, S. Tang, G. Wang, J. Bing, A. D. Bui, J. Qu, L. Yang, C. Liao, H. Chen, S. P. Bremner, H. T. Nguyen, J. Cairney, A. W. Y. Ho-Baillie, *Adv. Energy Mater.* **2022**, *12*, 2201672.
[17] J. Liu, M. De Bastiani, E. Aydin, G. T. Harrison, Y. Gao, R. R. Pradhan, M. K. Eswaran, M. Mandal, W. Yan, A. Seitkhan, M. Babics, A. S. Subbiah, E. Ugur, F. Xu, L. Xu, M. Wang, A. ur Rehman, A. Razzaq, J. Kang, R. Azmi, A. A. Said, F. H. Isikgor, T. G. Allen, D. Andrienko, U. Schwingenschlögl, F. Laquai, S. De Wolf, *Science* **2022**, *377*, 302.
[18] J. Lee, H. Kang, G. Kim, H. Back, J. Kim, S. Hong, B. Park, E. Lee, K. Lee, *Adv. Mater.* **2017**, *29*, 1606363.
[19] B. Li, Y. Xiang, K. D. G. I. Jayawardena, D. Luo, Z. Wang, X. Yang, J. F. Watts, S. Hinder, M. T. Sajjad, T. Webb, H. Luo, I. Marko, H. Li, S. A. J. Thomson, R. Zhu, G. Shao, S. J. Sweeney, S. R. P. Silva, W. Zhang, *Nano Energy* **2020**, *78*, 105249.
[20] B. Li, Y. Xiang, K. D. G. I. Jayawardena, D. Luo, J. F. Watts, S. Hinder, H. Li, V. Ferguson, H. Luo, R. Zhu, S. R. P. Silva, *Sol. RRL* **2020**, *4*, 2000060.
[21] M. Zhang, X. Yao, Z. Liu, in *2021 19th Int. Conf. on Optical Communications and Networks (ICOCN)*, IEEE, Qufu, China **2021**, pp. 1–3.
[22] M. Schultes, N. Giesbrecht, J. Küffner, E. Ahlswede, P. Docampo, T. Bein, M. Powalla, *ACS Appl. Mater. Interfaces* **2019**, *11*, 12948.
[23] R. Li, L. Cai, Y. Zou, H. Xu, Y. Tan, Y. Wang, J. Li, X. Wang, Y. Li, Y. Qin, D. Liang, T. Song, B. Sun, *ACS Appl. Mater. Interfaces* **2020**, *12*, 36681.
[24] J. You, F. Guo, S. Qiu, W. He, C. Wang, X. Liu, W. Xu, Y. Mai, *J. Energy Chem.* **2019**, *38*, 192.
[25] H. Ju, Y. Ma, Y. Cao, Z. Wang, L. Liu, M. Wan, T. Mahmoudi, Y.-B. Hahn, Y. Wang, Y. Mai, *Sol. RRL* **2022**, *6*, 2101082.
[26] D. Song, S. Narra, M. Y. Li, J. S. Lin, E. W. G. Diau, *ACS Energy Lett.* **2021**, *6*, 4179.
[27] K. Datta, J. Wang, D. Zhang, V. Zardetto, W. H. M. Remmerswaal, C. H. L. Weijtens, M. M. Wienk, R. A. J. Janssen, *Adv. Mater.* **2022**, *34*, 2110053.
[28] P. Tockhorn, J. Sutter, A. Cruz, P. Wagner, K. Jäger, D. Yoo, F. Lang, M. Grischek, B. Li, J. Li, O. Shargaieva, E. Unger, A. Al-Ashouri, E. Köhnen, M. Stolterfoht, D. Neher, R. Schlatmann, B. Rech, B. Stannowski, S. Albrecht, C. Becker, *Nat. Nanotechnol.* **2022**, *17*, 1214.
[29] A. Al-Ashouri, M. Marčinskas, E. Kasparavičius, T. Malinauskas, A. Palmstrom, V. Getautis, S. Albrecht, M. D. Mcgehee, A. Magomedov, *ACS Energy Lett.* **2023**, *8*, 898.
[30] W. Peng, K. Mao, F. Cai, H. Meng, Z. Zhu, T. Li, S. Yuan, Z. Xu, X. Feng, J. Xu, M. D. McGehee, J. Xu, *Science* **2023**, *379*, 683.
[31] H. Li, L. Zhang, H. Dai, H. He, *Inorg. Chem.* **2009**, *48*, 4421.
[32] M. M. Lee, J. Teuscher, T. Miyasaka, T. N. Murakami, H. J. Snaith, *Science* **2012**, *338*, 643.
[33] S. D. Stranks, G. E. Eperon, G. Grancini, C. Menelaou, M. J. P. Alcocer, T. Leijtens, L. M. Herz, A. Petrozza, H. J. Snaith, *Science* **2013**, *342*, 341.
[34] J. F. Joanny, P. G. de Gennes, *J. Chem. Phys.* **1984**, *81*, 552.
[35] R. Tadmor, *Langmuir* **2021**, *37*, 6357.
[36] H. Bin, K. Datta, J. Wang, T. P. A. Van Der Pol, J. Li, M. M. Wienk, R. A. J. Janssen, *ACS Appl. Mater. Interfaces* **2022**, *14*, 16497.
[37] N. Phung, M. Verheijen, A. Todinova, K. Datta, M. Verhage, A. Al-Ashouri, H. Köbler, X. Li, A. Abate, S. Albrecht, M. Creatore, *ACS Appl. Mater. Interfaces* **2022**, *14*, 2166.
[38] B. Wu, Q. Fu, L. Sun, Y. Liu, Z. Sun, S. Xue, Y. Liu, M. Liang, *ACS Energy Lett.* **2022**, *7*, 2667.
[39] Q. An, F. Paulus, D. Becker-koch, C. Cho, Q. Sun, A. Weu, S. Bitton, N. Tessler, Y. Vaynzof, *Matter* **2021**, *4*, 1683.
[40] T. I. Alanazi, O. S. Game, J. A. Smith, R. C. Kilbride, C. Greenland, R. Jayaprakash, K. Georgiou, N. J. Terrill, D. G. Lidzey, *RSC Adv.* **2020**, *10*, 40341.
[41] M. Saliba, T. Matsui, J. Y. Seo, K. Domanski, J. P. Correa-Baena, M. K. Nazeeruddin, S. M. Zakeeruddin, W. Tress, A. Abate, A. Hagfeldt, M. Grätzel, *Energy Environ. Sci.* **2016**, *9*, 1989.
[42] T. Du, C. H. Burgess, J. Kim, J. Zhang, J. R. Durrant, M. A. McLachlan, *SustainSustainable Energy Fuels* **2017**, *1*, 119.
[43] S. Wang, A. Wang, X. Deng, L. Xie, A. Xiao, C. Li, Y. Xiang, T. Li, L. Ding, F. Hao, *J. Mater. Chem. A* **2020**, *8*, 12201.
[44] B. Kasprzyk-Hordern, *Adv. Colloid Interface Sci.* **2004**, *110*, 19.
[45] E. B. Cornelius, *J. Phys. Chem.* **1955**, *59*, 809.
[46] S. Tan, T. Huang, Y. Yang, *J. Phys.: Energy* **2021**, *3*, 42003.
[47] T. Kirchartz, F. Deledalle, P. S. Tuladhar, J. R. Durrant, J. Nelson, *J. Phys. Chem. Lett.* **2013**, *4*, 2371.
[48] J. Chen, X. Xu, X. Tang, T. Jiang, Z. Chen, L. Ye, Y. Tian, X. Chen, T. Liu, Y. Su, H. Zhu, D. Di, L. Li, Y. M. Yang, *J. Phys. Chem. Lett.* **2021**, *12*, 5580.
[49] M. V. Khenkin, E. A. Katz, A. Abate, G. Bardizza, J. J. Berry, C. Brabec, F. Brunetti, V. Bulović, Q. Burlingame, A. Di Carlo, R. Cheacharoen, Y. B. Cheng, A. Colsmann, S. Cros, K. Domanski, M. Dusza, C. J. Fell, S. R. Forrest, Y. Galagan, D. Di Girolamo, M. Grätzel, A. Hagfeldt, E. von Hauff, H. Hoppe, J. Kettle, H. Köbler, M. S. Leite, S. F. Liu, Y. L. Loo, J. M. Luther, et al., *Nat. Energy* **2020**, *5*, 35.

